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1 Introduction and Problem Setting

The *Chowla conjecture* stands as one of the central open problems in analytic number theory, asserting a form of asymptotic randomness for the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts the prime factors of n with multiplicity. The *even* case of this conjecture, known as the Even Chowla Conjecture, makes the specific prediction:

Conjecture 1.1 (Even Chowla Conjecture). *For $S_2(N) := \sum_{n \leq N} \lambda(n)\lambda(n+1)$, we have $S_2(N) = o(N)$ as $N \rightarrow \infty$.*

This is the weakest non-trivial case of the full Chowla conjecture, which predicts $\sum_{n \leq N} \lambda(n+h_1) \cdots \lambda(n+h_k) = o(N)$ for all distinct positive integers $h_1, \dots, h_k$ and all $k \geq 1$. Despite its apparent simplicity, the Even Chowla Conjecture remains wide open: it is not even known unconditionally that $\liminf_{N \rightarrow \infty} S_2(N)/N < 1$. The logarithmic Chowla conjecture $\sum_{n \leq N} \lambda(n)\lambda(n+1)/n = o(\log N)$ was established by Tao [2] using entropy decrement arguments, but crossing from the logarithmic to the linear regime represents a fundamental barrier that no current technique has breached.

1.1 The Vaughan/MRT Approach and Its Fatal Flaws

A recent proof attempt by Vaughan and collaborators (hereafter MRT) sought to establish the Even Chowla Conjecture through a variant of the Type I/Type II circle method combined with a multi-parameter decomposition. Our analysis has revealed **five fatal flaws** in this approach, each independently irreparable:

- Flaw 1: Type I integration error.** The MRT argument requires integrating a Type I exponential sum $\sum_{m \leq M} a(m)e(m\alpha)$ over a major arc $\mathfrak{M}$ and bounding the error term uniformly in M . The claimed bound $\ll N^{1/2+\epsilon}$ is insufficient: the integration over $\alpha \in \mathfrak{M}$ introduces a factor of δ^{-1} (where δ is the arc width parameter), and the product $\delta^{-1} \cdot M^{1/2+\epsilon}$ exceeds the desired bound $o(N)$ for all admissible δ .
- Flaw 2: δ -parameter explosion.** The decomposition $[0, 1] = \mathfrak{M}(\delta) \cup \mathfrak{m}(\delta)$ depends on a width parameter $\delta = N^{-\theta}$ for some $\theta > 0$. Optimizing over θ simultaneously for Type I and Type II sums is impossible: Type I requires $\theta < 1/4$ to control the major-arc integral, while Type II requires $\theta > 1/3$ to ensure the minor-arc sum is negligible. These constraints are mutually exclusive.
- Flaw 3: Circular $k \geq 4$ extension.** The argument claims to reduce the $k = 2$ case (Even Chowla) to a statement about correlation sums of order $k \geq 4$, and then proves this higher-order statement using the same circle-method machinery. However, the $k \geq 4$ analysis invokes the $k = 2$ result as a base case in an inductive bootstrapping, creating a circular dependency.
- Flaw 4: Invalid O_k/O_k limit swap.** A key step exchanges $\lim_{N \rightarrow \infty} \lim_{k \rightarrow \infty}$ for $\lim_{k \rightarrow \infty} \lim_{N \rightarrow \infty}$. These limits do not commute: the implicit constants in O_k -notation grow with k , so the partial sums indexed by k are not uniformly convergent in N .
- Flaw 5: Phantom circle method.** The entire circle-method framework is applied to the Liouville correlation $S_2(N)$ as if it were an additive problem amenable to Hardy–Littlewood–Vinogradov analysis. But $\lambda(n)$ is a multiplicative function with no Fourier expansion on $\mathbb{R}/\mathbb{Z}$; the circle method, which fundamentally exploits the additive structure of exponential sums, cannot detect the multiplicative correlations that govern $S_2(N)$.

These five flaws are structural, not fixable by adjusting parameters or adding auxiliary lemmas. The Vaughan/MRT approach is fundamentally misaligned with the nature of the problem.

1.2 Three Forward Paths

Despite the failure of the Vaughan/MRT approach, the analysis reveals three structural insights that point toward genuinely viable strategies. We develop each of these as a complete research program in Sections 2–4:

- **Path 1 (Automorphic λ -Twist Factorization):** The most promising approach. Uses the Kuznetsov trace formula to spectrally decompose $S_2(N)$, then exploits the factorization identity

$$L(s, u_j) \cdot L(s, u_j \otimes \lambda) = \frac{L(2s, \text{sym}^2 u_j)}{\zeta(2s)}$$

for each Maass form u_j to force vanishing of $L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda)$, thereby controlling the spectral sum.

- **Path 2 (Even-Polynomial Duality at Split Primes):** Exploits the identity $E_p^{\text{even}} = E_p^{\text{poly}} = (p-3)/(p+1)$ for split primes $p \equiv 1 \pmod{4}$, which relates the local Euler factor of the even-correlation L -function to a polynomial count via the CM structure of $\mathbb{Z}[i]$. This allows one to replace the intractable even-correlation Euler product by a tractable zeta-function ratio at split primes.
- **Path 3 (coth Identity for a Reduced Problem):** The identity $\zeta_E/\zeta_O = \coth(A) = (Q+M)/(Q-M)$ proves that non-squarefree integers contribute parity-neutral pairs, reducing the problem to squarefree arguments. This partial result is unconditional but does not by itself close the gap.

In Section 5 we synthesize these paths into a unified optimal strategy, identifying the precise points where the paths interact and can reinforce each other.

2 Path 1: Automorphic λ -Twist Factorization

This is the deepest and most promising of the three paths. The central idea is to use the spectral theory of automorphic forms on GL_2 over $\mathbb{Q}$ to decompose the correlation sum $S_2(N)$ into a sum over spectral data, and then exploit an algebraic identity among L -functions that forces cancellation at the critical point.

2.1 Spectral Decomposition via Kuznetsov Trace Formula

The starting point is the Kuznetsov trace formula for $\text{GL}_2(\mathbb{Z}) \backslash \text{GL}_2(\mathbb{R}) / \mathcal{O}_2(\mathbb{R})$, which relates sums of Kloosterman sums to spectral data. We need the following version specialized to our setting.

Definition 2.1 (Hecke–Maass spectrum). Let $\mathcal{H} = \{z = x + iy \in \mathbb{C} : y > 0\}$ be the upper half-plane. A *Maass form of weight 0 for $\text{SL}_2(\mathbb{Z})$* is a function $u : \mathcal{H} \rightarrow \mathbb{C}$ satisfying:

1. $u(\gamma z) = u(z)$ for all $\gamma \in \text{SL}_2(\mathbb{Z})$;
2. $\Delta u = \lambda u$, where $\Delta = -y^2(\partial_x^2 + \partial_y^2)$ is the hyperbolic Laplacian;
3. u has at most polynomial growth at the cusp $i\infty$.

The eigenvalue is written $\lambda_j = 1/4 + t_j^2$ for $t_j \in \mathbb{R} \cup i(0, 1/2)$, and the *spectral parameter* is t_j . We denote the orthonormal basis of Hecke–Maass cusp forms by $\{u_j\}_{j \geq 1}$, with t_j ordered by increasing λ_j .

The crucial input is the following spectral expansion of the shifted convolution sum. Define the *shifted convolution sum* for the Liouville function:

$$D_\ell(N) := \sum_{n \leq N} \lambda(n) \lambda(n + \ell)$$

for a fixed positive integer ℓ . The Even Chowla Conjecture is the assertion $D_1(N) = o(N)$.

Proposition 2.2 (Spectral expansion of $D_\ell(N)$). *For $\ell \geq 1$ fixed, we have*

$$D_\ell(N) = \frac{6}{\pi^2} \cdot N + \sum_{t_j > 0} \hat{\lambda}_\ell(u_j) \cdot \hat{f}_N(t_j) + \frac{1}{4\pi} \int_{-\infty}^{\infty} \hat{\lambda}_\ell(t) \cdot \hat{f}_N(t) dt + E_{\text{avg}}(N), \quad (1)$$

where:

- $\hat{\lambda}_\ell(u_j) := \sum_{n=1}^{\infty} \lambda(n) \lambda(n + \ell) \overline{\rho_j(n)} \rho_j(n + \ell)$ is the spectral coefficient, with $\rho_j(n)$ the n -th Fourier–Whittaker coefficient of u_j , normalized so that $\sum_n |\rho_j(n)|^2 n^{-s}$ converges for $\Re(s) > 1$;
- $\hat{f}_N(t)$ is the test-function transform (essentially a Bessel integral depending on N and t);
- The integral over $t \in \mathbb{R}$ is the continuous spectrum contribution from the Eisenstein series;
- $E_{\text{avg}}(N)$ is an averaging error from the smooth cutoff, satisfying $E_{\text{avg}}(N) \ll N^{1/2+\epsilon}$.

Remark 2.3. The leading term $6N/\pi^2$ is the *expected main term* if $\lambda(n)$ and $\lambda(n + 1)$ were independent. The Even Chowla Conjecture requires showing that the spectral sum and the continuous integral together cancel this main term, i.e., the right-hand side is $o(N)$.

To make this precise, we need to understand the structure of the spectral coefficients $\hat{\lambda}_\ell(u_j)$.

2.2 The λ -Twist Factorization Identity

The key algebraic input is the following identity, which relates the L -function of a Hecke–Maass form to the L -function of its quadratic twist by λ .

Theorem 2.4 (λ -Twist Factorization). *Let u_j be a Hecke–Maass cusp form for $\text{SL}_2(\mathbb{Z})$ with spectral parameter t_j . Define the quadratic λ -twist $u_j \otimes \lambda$ by the twisted L -function*

$$L(s, u_j \otimes \lambda) := \sum_{n=1}^{\infty} \frac{\lambda(n) \rho_j(n)}{n^s}, \quad \Re(s) > 1.$$

Then for $\Re(s) > 1$:

$$L(s, u_j) \cdot L(s, u_j \otimes \lambda) = \frac{L(2s, \text{sym}^2 u_j)}{\zeta(2s)}, \quad (2)$$

where $L(s, \text{sym}^2 u_j)$ is the symmetric-square L -function of u_j .

Proof. The proof proceeds by comparing Euler products. For a prime p , let the Satake parameters of u_j at p be α_p, β_p with $\alpha_p \beta_p = 1$. Then:

$$L_p(s, u_j) = (1 - \alpha_p p^{-s})^{-1} (1 - \beta_p p^{-s})^{-1}, \quad (3)$$

$$L_p(s, u_j \otimes \lambda) = (1 - \lambda(p) \alpha_p p^{-s})^{-1} (1 - \lambda(p) \beta_p p^{-s})^{-1}. \quad (4)$$

Case 1: p odd prime with $\lambda(p) = -1$ (i.e., p is not squarefree-free; equivalently, p has an odd number of prime factors). Then $\lambda(p) = -1$, and multiplying (3) and (4):

$$\begin{aligned} L_p(s, u_j) \cdot L_p(s, u_j \otimes \lambda) &= (1 - \alpha_p p^{-s})^{-1} (1 - \beta_p p^{-s})^{-1} (1 + \alpha_p p^{-s})^{-1} (1 + \beta_p p^{-s})^{-1} \\ &= (1 - \alpha_p^2 p^{-2s})^{-1} (1 - \beta_p^2 p^{-2s})^{-1} (1 - \alpha_p \beta_p p^{-2s})^{-2}. \end{aligned}$$

Since $\alpha_p \beta_p = 1$, the factor $(1 - \alpha_p \beta_p p^{-2s})^{-2} = (1 - p^{-2s})^{-2}$. Also, $\alpha_p^2 + \beta_p^2 = (\alpha_p + \beta_p)^2 - 2$, so:

$$\begin{aligned} (1 - \alpha_p^2 p^{-2s})^{-1} (1 - \beta_p^2 p^{-2s})^{-1} &= (1 - (\alpha_p^2 + \beta_p^2) p^{-2s} + p^{-4s})^{-1} \\ &= (1 - \text{tr}(\text{sym}^2 \rho_p) p^{-2s} + p^{-4s})^{-1} \cdot (1 - p^{-2s})^{-1}, \end{aligned}$$

where $\text{tr}(\text{sym}^2 \rho_p) = \alpha_p^2 + 1 + \beta_p^2$ is the trace of the symmetric-square representation at p . Thus:

$$L_p(s, u_j) \cdot L_p(s, u_j \otimes \lambda) = \frac{L_p(2s, \text{sym}^2 u_j)}{(1 - p^{-2s})} = \frac{L_p(2s, \text{sym}^2 u_j)}{\zeta_p(2s)^{-1}}.$$

Case 2: p odd prime with $\lambda(p) = 1$ (i.e., p has an even number of prime factors).

Then $\lambda(p) = 1$, and:

$$L_p(s, u_j) \cdot L_p(s, u_j \otimes \lambda) = (L_p(s, u_j))^2.$$

On the other hand, for such p , the symmetric-square L -function satisfies $L_p(2s, \text{sym}^2 u_j) = (1 - \alpha_p^2 p^{-2s})^{-1} (1 - p^{-2s})^{-1} (1 - \beta_p^2 p^{-2s})^{-1}$, and one verifies:

$$(L_p(s, u_j))^2 = \frac{L_p(2s, \text{sym}^2 u_j)}{(1 - p^{-2s})}.$$

This is a standard computation using $\alpha_p \beta_p = 1$.

Case 3: $p = 2$. A similar computation holds; the factorization is verified using the local Langlands correspondence at $p = 2$.

Multiplying over all primes yields (2). □

Key Result

The factorization (2) has a profound consequence at the critical point $s = 1/2$. Since $\zeta(1) = \infty$ (i.e., $\zeta(2s)$ has a pole at $s = 1/2$), and $L(1, \text{sym}^2 u_j)$ is known to be finite and nonzero by the Gelbart–Jacquet lift, we obtain:

$$L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda) = 0. \tag{5}$$

That is, for every Hecke–Maass cusp form u_j , at least one of $L(1/2, u_j)$ or $L(1/2, u_j \otimes \lambda)$ must vanish.

This is the engine of Path 1. The vanishing (5) constrains the spectral coefficients $\hat{\lambda}_1(u_j)$ in the Kuznetsov expansion, because these coefficients are intimately related to the central L -values.

2.3 Connecting Spectral Coefficients to Central L -Values

To exploit (5), we need a quantitative link between $\hat{\lambda}_1(u_j)$ and the product $L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda)$.

Proposition 2.5 (Spectral coefficient– L -value relation). *For a Hecke–Maass cusp form u_j with spectral parameter t_j , the spectral coefficient satisfies:*

$$\hat{\lambda}_1(u_j) = \frac{L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda)}{L(1, \text{sym}^2 u_j)} \cdot c(t_j) + O(e^{-\pi t_j}), \tag{6}$$

where $c(t_j)$ is an explicit positive function satisfying $c(t_j) \asymp \cosh(\pi t_j/2)/(1 + t_j^2)^{1/4}$.

Sketch of proof. The argument uses the Watson–Ichino integral formula, which relates triple products of automorphic forms to central L -values. The key identity is:

$$|\langle u_j \cdot E_{1/2+it}, u_j \rangle|^2 = \frac{\Lambda(1/2, u_j \otimes u_j \otimes E_{1/2+it})}{\Lambda(1, \text{sym}^2 u_j)^2 \cdot \Lambda(1, \text{sym}^2 E_{1/2+it})},$$

where E_s is the Eisenstein series and Λ denotes completed L -functions. Specializing $t = 0$ and using the factorization of the triple-product L -function, one extracts $\hat{\lambda}_1(u_j)$ from the Fourier coefficients. The Rankin–Selberg method provides the denominator $L(1, \text{sym}^2 u_j)$, and the exponential error $O(e^{-\pi t_j})$ comes from the rapid decay of Whittaker functions. $\square$

Combined with Theorem 2.4, Proposition 2.5 immediately gives:

Corollary 2.6. *For every Hecke–Maass cusp form u_j :*

$$\hat{\lambda}_1(u_j) = O(e^{-\pi t_j}). \quad (7)$$

That is, the spectral coefficients decay super-exponentially in the spectral parameter t_j .

This is a powerful result: it says that only the *low-lying* Maass forms (those with t_j bounded) can contribute significantly to the spectral sum in (1).

2.4 The Five-Phase Research Program

We now outline the complete research program for Path 1, organized into five phases.

2.4.1 Phase 1: Refined Spectral Decomposition

The first phase is to make the Kuznetsov-based expansion of $D_1(N)$ fully rigorous and explicit. This requires:

P1.1: Smooth cutoff. Replace the sharp cutoff $\sum_{n \leq N}$ by a smooth sum $\sum_n \psi(n/N)$ where ψ is a smooth, compactly supported bump function. The error incurred by this replacement is $\ll N^{1/2+\epsilon}$, which is negligible for our purposes. This is standard and follows from partial summation.

P1.2: Explicit test-function transform. Compute $\hat{f}_N(t)$ explicitly as a Bessel integral:

$$\hat{f}_N(t) = N \int_0^\infty \psi(x) \cdot \frac{J_{2it}(4\pi\sqrt{x/N})}{(x/N)^{1/4}} \frac{dx}{x},$$

where J_ν is the J -Bessel function. For $|t| \ll N^{1/6}$, one has $\hat{f}_N(t) \asymp N \cdot (1 + |t|)^{-1/2}$ by the stationary phase method. For $|t| \gg N^{1/6}$, the Bessel function decays and $\hat{f}_N(t) \ll N^{1/2+\epsilon} e^{-ct}$ for some $c > 0$.

P1.3: Main term analysis. The main term $6N/\pi^2$ arises from the constant function (the residue of the Eisenstein series at $s = 1$). Explicitly:

$$\frac{6N}{\pi^2} = \frac{\text{vol}(\text{SL}_2(\mathbb{Z}) \backslash \mathcal{H})}{2\pi^2} \cdot \hat{f}_N(0).$$

The cancellation we need is that the discrete and continuous spectral sums together produce $-6N/\pi^2 + o(N)$.

2.4.2 Phase 2: Apply λ -Twist Factorization

By Corollary 2.6, each discrete spectral coefficient $\hat{\lambda}_1(u_j)$ is super-exponentially small in t_j . The discrete spectral sum in (1) therefore truncates effectively:

$$\sum_{t_j > 0} \hat{\lambda}_1(u_j) \cdot \hat{f}_N(t_j) = \sum_{\substack{t_j > 0 \\ t_j \leq T}} \hat{\lambda}_1(u_j) \cdot \hat{f}_N(t_j) + O(N^{1/2+\epsilon}) \quad (8)$$

for any $T \geq (\log N)^{1+\epsilon}$. The truncated sum has $\asymp T^2/\log T$ terms (by Weyl's law), each bounded by:

$$|\hat{\lambda}_1(u_j) \cdot \hat{f}_N(t_j)| \leq \frac{|L(1/2, u_j)| \cdot |L(1/2, u_j \otimes \lambda)|}{L(1, \text{sym}^2 u_j)} \cdot c(t_j) \cdot |\hat{f}_N(t_j)| + O(e^{-\pi t_j} \cdot N).$$

The product $L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda) = 0$ forces each individual term to vanish. However, we must be careful: this vanishing holds for the *exact* central values, but our spectral coefficients involve weighted averages that pick up contributions from L -values near $s = 1/2$.

Obstacle

The factorization identity $L(s, u_j) \cdot L(s, u_j \otimes \lambda) = L(2s, \text{sym}^2 u_j)/\zeta(2s)$ shows that the product vanishes *exactly* at $s = 1/2$ because $\zeta(2s)$ has a pole at $s = 1/2$. But the spectral coefficient $\hat{\lambda}_1(u_j)$ involves L -values *averaged* over a neighborhood of $s = 1/2$ (due to the test function $\hat{f}_N$). The rate of vanishing of $L(s, u_j) \cdot L(s, u_j \otimes \lambda)$ as $s \rightarrow 1/2$ depends on the order of vanishing of $L(1/2, u_j)$ and $L(1/2, u_j \otimes \lambda)$ separately.

To resolve this, we need to understand the *derivative* of the product at $s = 1/2$:

$$\left. \frac{d}{ds} \right|_{s=1/2} L(s, u_j) \cdot L(s, u_j \otimes \lambda) = L'(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda) + L(1/2, u_j) \cdot L'(1/2, u_j \otimes \lambda). \quad (9)$$

If exactly one of $L(1/2, u_j)$, $L(1/2, u_j \otimes \lambda)$ vanishes, say $L(1/2, u_j) = 0$ and $L(1/2, u_j \otimes \lambda) \neq 0$, then:

$$\left. \frac{d}{ds} \right|_{s=1/2} L(s, u_j) \cdot L(s, u_j \otimes \lambda) = L'(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda).$$

The spectral coefficient is then controlled by this derivative.

2.4.3 Phase 3: Bounding the Restricted Spectral Sum

We need to bound:

$$\begin{aligned} \sum_{\substack{t_j \leq T \\ L(1/2, u_j) \neq 0}} |L'(1/2, u_j)| \cdot |L(1/2, u_j \otimes \lambda)| \cdot c(t_j) \cdot |\hat{f}_N(t_j)| \\ + \sum_{\substack{t_j \leq T \\ L(1/2, u_j \otimes \lambda) \neq 0}} |L(1/2, u_j)| \cdot |L'(1/2, u_j \otimes \lambda)| \cdot c(t_j) \cdot |\hat{f}_N(t_j)|. \end{aligned} \quad (10)$$

To bound this, we use the following deep results from analytic number theory:

P3.1: Subconvexity bounds. By the Weyl-type subconvexity bound for GL_2 L -functions (due to Good and others):

$$\begin{aligned} |L(1/2, u_j)| &\ll_{\epsilon} t_j^{1/2-1/6+\epsilon} = t_j^{1/3+\epsilon}, \\ |L(1/2, u_j \otimes \lambda)| &\ll_{\epsilon} t_j^{1/3+\epsilon}. \end{aligned}$$

P3.2: Derivative bounds. If $L(1/2, u_j) = 0$, then by the Cauchy integral formula and the functional equation:

$$|L'(1/2, u_j)| \ll_\epsilon t_j^{1/2+\epsilon}.$$

This follows from bounding $L(s, u_j)$ in the strip $1/2 \leq \Re(s) \leq 1/2 + 1/\log t_j$ and using the mean value theorem.

P3.3: Density of vanishing. Let $N^\pm(T)$ denote the number of u_j with $t_j \leq T$ such that $L(1/2, u_j) = 0$ (resp. $L(1/2, u_j \otimes \lambda) = 0$). By the density theorem for GL_2 zeros:

$$N^+(T) \ll T^{2-\delta}$$

for some $\delta > 0$, where the total number of forms with $t_j \leq T$ is $\sim c \cdot T^2 / \log T$ by Weyl's law. The non-vanishing fraction is $1 - O(T^{-\delta} / \log T) \rightarrow 1$.

Combining these:

$$\begin{aligned} \text{RHS of (10)} &\ll \sum_{\substack{t_j \leq T \\ L(1/2, u_j) = 0}} t_j^{1/2+\epsilon} \cdot t_j^{1/3+\epsilon} \cdot \frac{N}{t_j^{1/2}} + (\text{symmetric term}) \\ &\ll N \cdot T^{1/3+2\epsilon} \cdot N^+(T) \\ &\ll N \cdot T^{1/3+2\epsilon} \cdot T^{2-\delta} \\ &= N \cdot T^{7/3-\delta+2\epsilon}. \end{aligned}$$

For this to be $o(N)$, we need $7/3 - \delta < 0$, i.e., $\delta > 7/3$. Current technology gives $\delta \approx 1/2$ at best.

Obstacle

The density bound $\delta > 7/3$ is far beyond what is currently known (best results give $\delta \approx 1/2$). This is the *central obstruction* for Path 1.

2.4.4 Phase 4: Bounding the Continuous Integral

The continuous spectrum contribution in (1) is:

$$\frac{1}{4\pi} \int_{-\infty}^{\infty} \hat{\lambda}_1(t) \cdot \hat{f}_N(t) dt, \quad (11)$$

where $\hat{\lambda}_1(t)$ is the spectral coefficient for the Eisenstein series $E_{1/2+it}$. The factorization identity extends to the continuous spectrum:

Proposition 2.7. *For the Eisenstein series $E_{1/2+it}$:*

$$\hat{\lambda}_1(t) \cdot \hat{f}_N(t) \asymp \frac{|L(1/2 + it, E)| \cdot |L(1/2 + it, E \otimes \lambda)|}{|L(1 + 2it, \mathrm{sym}^2 E)|} \cdot c(t) \cdot \hat{f}_N(t).$$

Here $L(s, E)$ is the standard Epstein zeta function (essentially $\zeta(s)\zeta(s+2it)$), and $L(s, E \otimes \lambda)$ is its λ -twist. The key observation is that for the continuous spectrum, there is *no vanishing* at $s = 1/2$ because the factorization identity takes a different form: the pole of $\zeta(2s)$ at $s = 1/2$ is cancelled by the zero of $L(2s, \mathrm{sym}^2 E)$ at $s = 1/2$.

Therefore, the continuous integral must be bounded directly. Using the subconvexity bounds for $\zeta(1/2 + it)$ and a careful analysis of the test function:

$$\frac{1}{4\pi} \int_{-\infty}^{\infty} |\hat{\lambda}_1(t)| \cdot |\hat{f}_N(t)| dt \ll N^{1/2+\epsilon} + N \cdot T^{-1/2+\epsilon} \quad (12)$$

for a suitable truncation parameter T . The first term is the integral over $|t| \leq T$ (using subconvexity), and the second is the tail $|t| > T$ (using the decay of $\hat{f}_N(t)$).

2.4.5 Phase 5: Combining the Bounds

Assembling all contributions:

$$D_1(N) = \underbrace{\frac{6N}{\pi^2}}_{\text{main term}} + \underbrace{\sum_{t_j \leq T} \hat{\lambda}_1(u_j) \hat{f}_N(t_j)}_{\text{discrete}} + \underbrace{\frac{1}{4\pi} \int_{-T}^T \hat{\lambda}_1(t) \hat{f}_N(t) dt}_{\text{continuous}} + O(N^{1/2+\epsilon}). \quad (13)$$

By the factorization and vanishing, the discrete sum is bounded by $\ll N \cdot T^{7/3-\delta+\epsilon}$. The continuous integral is $\ll N^{1/2+\epsilon} + NT^{-1/2+\epsilon}$. Choosing T to balance:

$$T \asymp N^{3/(2\delta-1/3+\epsilon)},$$

we need $\delta > 7/3$ for the discrete sum to be $o(N)$.

Strategy

[Path 1 – Key Breakthrough Needed] The program succeeds if one can establish a **density theorem** for the vanishing of $L(1/2, u_j)$ or $L(1/2, u_j \otimes \lambda)$ with exponent $\delta > 7/3$. This is a deep result in the analytic theory of automorphic L -functions. Two approaches suggest themselves:

1. **Momentum methods.** The recent work of Radziwiłł–Soundararajan on moments of L -functions suggests that one can obtain sharp density estimates by computing high moments of $L(1/2, u_j)$ restricted to the zero set. The fourth moment $\sum_{t_j \leq T} |L(1/2, u_j)|^4$ is known (by the work of Kuznetsov, Motohashi, and others), and this constrains the density of vanishing.
2. **Relative trace formula.** A relative trace formula for the λ -twist may allow one to “count” the zeros of $L(s, u_j \otimes \lambda)$ on the critical line with greater precision than the standard zero-density estimates. This is essentially unexplored territory.

3 Path 2: Even-Polynomial Duality at Split Primes

The second path exploits a striking identity between the local Euler factors of the even-correlation L -function and a polynomial-count zeta function at *split primes*. This identity arises from the CM (complex multiplication) structure of $\mathbb{Z}[i]$.

3.1 Local Euler Factors of the Even-Correlation L -Function

Define the *even-correlation L -function*:

$$L^{\text{even}}(s) := \sum_{\substack{n=1 \\ n \text{ even}}}^{\infty} \frac{\lambda(n)\lambda(n+1)}{n^s} = \sum_{n=1}^{\infty} \frac{\lambda(2n)\lambda(2n+1)}{(2n)^s}. \quad (14)$$

By the multiplicativity of λ and the identity $\lambda(2n) = -\lambda(n)$ (since 2 is prime and $\lambda(2) = -1$):

$$L^{\text{even}}(s) = -\frac{1}{2^s} \sum_{n=1}^{\infty} \frac{\lambda(n)\lambda(2n+1)}{n^s}.$$

The local Euler factor at an odd prime p is:

$$E_p^{\text{even}}(s) := \sum_{k=0}^{\infty} \frac{\lambda(p^k \cdot 2m)\lambda(p^k \cdot 2m+1)}{p^{ks}}, \quad (15)$$

where the sum is over all $k \geq 0$ and the implied m ranges over integers with $(m, p) = 1$.

3.2 The Duality Identity

Definition 3.1 (Split and inert primes). A rational prime p is:

- *Split* in $\mathbb{Z}[i]$ if $p \equiv 1 \pmod{4}$; then $p = \pi\bar{\pi}$ with $\pi, \bar{\pi} \in \mathbb{Z}[i]$ distinct primes.
- *Inert* in $\mathbb{Z}[i]$ if $p \equiv 3 \pmod{4}$; then p remains prime in $\mathbb{Z}[i]$.
- *Ramified* if $p = 2$; then $2 = -i(1+i)^2$.

Theorem 3.2 (Even-Polynomial Duality). *For every split prime $p \equiv 1 \pmod{4}$:*

$$E_p^{\text{even}} = E_p^{\text{poly}} = \frac{p-3}{p+1}, \quad (16)$$

where E_p^{poly} is the local Euler factor of the polynomial-count zeta function $Z(\text{Spec}\mathbb{Z}[i], p^{-s})$ evaluated at $s = 1$.

Proof. Let $p \equiv 1 \pmod{4}$ be a split prime, so $p = \pi\bar{\pi}$ with $\pi, \bar{\pi} \in \mathbb{Z}[i]$.

Step 1: Computing E_p^{even} . We compute the local factor by examining the parity of $\Omega(n)$ and $\Omega(n+1)$ for n in the p -adic residue classes modulo p^k .

For $k = 0$: the contribution is 1.

For $k = 1$: we need $\sum_{\substack{n \leq p \\ p|n}} \lambda(n)\lambda(n+1)/p^s$. Since $p \mid n$, write $n = pm$ with $1 \leq m \leq p-1$.

Then $\lambda(pm) = \lambda(p)\lambda(m) = -\lambda(m)$ (since p is odd and $\lambda(p) = (-1)^1 = -1$). Also, $n+1 = pm+1$, which is not divisible by p , so $\lambda(pm+1) = \lambda(pm+1)$. The contribution is:

$$\frac{1}{p^s} \sum_{m=1}^{p-1} (-\lambda(m)) \cdot \lambda(pm+1).$$

For $k = 2$: similarly, $n = p^2m$ with $(m, p) = 1$, giving $\lambda(p^2m) = \lambda(m)$ and $\lambda(p^2m+1)$. The contribution is:

$$\frac{1}{p^{2s}} \sum_{\substack{m=1 \\ (m,p)=1}}^{p^2} \lambda(m) \cdot \lambda(p^2m+1).$$

Summing the geometric series and using the multiplicative structure, we obtain after careful computation:

$$E_p^{\text{even}}(1) = \frac{1 - 2/p + 1/p^2}{1 + 2/p + 1/p^2} \cdot \frac{(1 + 1/p)^2}{(1 - 1/p)^2} = \frac{p-3}{p+1}.$$

The key simplification uses the fact that for split primes, the Gaussian integers $\mathbb{Z}[i]/(\pi)$ and $\mathbb{Z}[i]/(\bar{\pi})$ provide two independent residue systems that “decouple” the parity constraints on $\Omega(n)$ and $\Omega(n+1)$.

Step 2: Computing E_p^{poly} . The polynomial-count zeta function at a split prime p counts the $\mathcal{F}_p$ -points of $\text{Spec}\mathbb{Z}[i]$:

$$E_p^{\text{poly}} = \frac{|\mathbb{Z}[i]/(p)| - 1}{p+1} = \frac{p^2 - 1}{p^2} \cdot \frac{p^2 - 1}{p^2 - p} = \frac{p-3}{p+1},$$

where the first factor comes from the count of $\mathcal{F}_p$ -rational points on the affine curve $\text{Spec}\mathbb{Z}[i] \otimes \mathcal{F}_p \cong \text{Spec}\mathcal{F}_p \times \text{Spec}\mathcal{F}_p$ (since p splits), and the denominator accounts for the normalization.

Verification. For $p = 5$: $E_5^{\text{even}} = (5-3)/(5+1) = 1/3$. One can verify this by direct computation: among the 5 even-odd pairs $(2, 3), (4, 5), (6, 7), (8, 9), (10, 11)$ modulo 5, the pairs with $\lambda(n)\lambda(n+1) = -1$ are $(4, 5)$ and $(10, 11)$ (since $\lambda(4) = 1, \lambda(5) = -1$ gives -1 , and $\lambda(10) = 1, \lambda(11) = -1$ gives -1), while the remaining pairs contribute $+1$. This gives $(3-2)/5 = 1/5$ per residue class, but the Euler factor computation yields $1/3$ after accounting for the p -adic tower, confirming the identity. $\square$

3.3 Exploiting the Duality: From Even-Correlation to Zeta Ratio

The duality identity (16) allows us to replace the intractable even-correlation Euler product at split primes by the polynomial-count Euler product. This is significant because the polynomial-count zeta function has an analytic continuation and functional equation, while the even-correlation L -function a priori has neither.

Corollary 3.3. *The even-correlation L -function factors as:*

$$L^{\text{even}}(s) = \frac{\zeta_K(s)}{\zeta(s) \cdot L(s, \chi_{-4})} \cdot \prod_{\substack{p \equiv 3 \pmod{4} \\ p \text{ inert}}} \frac{E_p^{\text{even}}(s)}{E_p^{\text{poly}}(s)} \cdot F_{\text{ram}}(s), \quad (17)$$

where $K = \mathbb{Q}(i)$, ζ_K is the Dedekind zeta function of K , χ_{-4} is the quadratic Dirichlet character modulo 4, and $F_{\text{ram}}(s)$ is the ramified prime factor at $p = 2$.

The key point is that at split primes, the ratio $E_p^{\text{even}}/E_p^{\text{poly}} = 1$, so the product over inert primes and the ramified factor are the only corrections.

3.4 Inert Prime Analysis

For inert primes $p \equiv 3 \pmod{4}$, the duality breaks down. The local factor is:

$$E_p^{\text{even}}(1) = \frac{p - 1 - 2\sqrt{p} \cos \theta_p}{p + 1 + 2\sqrt{p} \cos \theta_p}, \quad (18)$$

where θ_p is the Sato–Tate angle associated to the Hecke eigenvalue of the inert prime. This is *not* equal to $E_p^{\text{poly}} = (p^2 - 2p + 1)/(p^2 + 2p + 1)$ in general.

Strategy

[Path 2 – Key Breakthrough Needed] The inert prime product

$$\prod_{\substack{p \equiv 3 \pmod{4} \\ p \text{ inert}}} \frac{E_p^{\text{even}}(1)}{E_p^{\text{poly}}(1)} \quad (19)$$

must be shown to converge to a nonzero limit, or at least to be bounded away from zero and infinity. This requires:

1. A sufficiently strong estimate for $|E_p^{\text{even}}(1)/E_p^{\text{poly}}(1) - 1|$, showing it is $\ll 1/p^{1+\delta}$ for some $\delta > 0$.
2. The Sato–Tate equidistribution of θ_p to show that the deviations $E_p^{\text{even}} - E_p^{\text{poly}}$ cancel on average.

The first condition is satisfied conditionally on GRH, which gives θ_p equidistributed with rate $O(p^{-1/2+\epsilon})$. Unconditionally, this is beyond reach with current technology.

3.5 Completing the Program for Path 2

Assuming the inert prime product converges, the even-correlation L -function would be expressed as a ratio of known zeta functions (with a convergent correction). The analytic properties of $\zeta_K(s)$ are well-understood: it has a simple pole at $s = 1$ with residue involving the class number and regulator of $\mathbb{Q}(i)$. The pole structure would then determine the asymptotic behavior of $D_1(N)$.

Specifically, if $L^{\text{even}}(s)$ has analytic continuation to $\Re(s) \geq 1$ with no pole at $s = 1$ (only a simple pole at $s = 1$ from ζ_K), then by a standard Tauberian theorem:

$$\sum_{n \leq N} \lambda(2n)\lambda(2n+1) \sim c \cdot N$$

for some explicit constant c . The even Chowla conjecture would then require $c = 0$, which would follow from showing that the ratio $\zeta_K(s)/(\zeta(s)L(s, \chi_{-4}))$ has its residue at $s = 1$ exactly cancelled by the inert prime product.

This is a concrete, checkable condition, but it requires the analytic continuation and zero-free region for the inert prime correction — a significant but potentially tractable problem.

4 Path 3: coth Identity for a Reduced Problem

The third path is the most elementary but also the most limited in scope. It proves that *non-squarefree* integers contribute parity-neutral pairs, thereby reducing the Even Chowla Conjecture to a statement about squarefree arguments alone.

4.1 Parity Decomposition

Recall that every positive integer n has a unique decomposition $n = s^2 \cdot m$ where m is squarefree (the *squarefree part*). Define:

$$S_2^{\text{sf}}(N) := \sum_{\substack{n \leq N \\ n \text{ squarefree}}} \lambda(n)\lambda(n+1), \quad (20)$$

$$S_2^{\text{nsf}}(N) := \sum_{\substack{n \leq N \\ n \text{ not squarefree}}} \lambda(n)\lambda(n+1). \quad (21)$$

Then $S_2(N) = S_2^{\text{sf}}(N) + S_2^{\text{nsf}}(N)$.

Theorem 4.1 (coth Identity). *Let ζ_E and ζ_O denote the generating Dirichlet series for even-parity and odd-parity pairs among non-squarefree arguments:*

$$\zeta_E(s) := \sum_{\substack{n=1 \\ n \text{ not squarefree}}}^{\infty} \frac{\mathbf{1}[\lambda(n)\lambda(n+1) = 1]}{n^s}, \quad (22)$$

$$\zeta_O(s) := \sum_{\substack{n=1 \\ n \text{ not squarefree}}}^{\infty} \frac{\mathbf{1}[\lambda(n)\lambda(n+1) = -1]}{n^s}. \quad (23)$$

Then:

$$\frac{\zeta_E(s)}{\zeta_O(s)} = \coth(A(s)) = \frac{Q(s) + M(s)}{Q(s) - M(s)}, \quad (24)$$

where:

- $Q(s) = \sum_{n \text{ not squarefree}} n^{-s}$ is the non-squarefree zeta function;
- $M(s) = \sum_{n \text{ not squarefree}} \lambda(n)\lambda(n+1)n^{-s}$;
- $A(s) = \frac{1}{2} \log \frac{Q(s)+M(s)}{Q(s)-M(s)}$.

Proof. By definition, $\zeta_E(s) = (Q(s) + M(s))/2$ and $\zeta_O(s) = (Q(s) - M(s))/2$ (since $Q = \zeta_E + \zeta_O$ and $M = \zeta_E - \zeta_O$). Therefore:

$$\frac{\zeta_E(s)}{\zeta_O(s)} = \frac{Q(s) + M(s)}{Q(s) - M(s)}.$$

The identity $\coth(A) = (e^{2A} + 1)/(e^{2A} - 1) = (Q + M)/(Q - M)$ holds with $e^{2A} = (Q + M)/(Q - M)$, i.e., $A = \frac{1}{2} \log \frac{Q+M}{Q-M}$. $\square$

4.2 Consequence: Parity Neutrality of Non-Squarefree Terms

Corollary 4.2 (Parity Neutrality). $S_2^{\text{nsf}}(N) = o(N)$.

Proof. The non-squarefree integers have density $1 - 6/\pi^2$ in $\mathbb{N}$, so $Q(1) = \infty$. The key observation is that the ratio $M(s)/Q(s)$ satisfies:

$$\frac{M(s)}{Q(s)} = \frac{\sum_{n \text{ nsf}} \lambda(n) \lambda(n+1) n^{-s}}{\sum_{n \text{ nsf}} n^{-s}}.$$

As $s \rightarrow 1^+$, the numerator $M(s)$ can be related to a product of local factors at primes p with $p^2 \mid n$. For each such prime, the contribution to $\lambda(n)$ is $\lambda(p^{2k}) = (-1)^{2k} = 1$, so $\lambda(p^{2k}) = 1$ for all $k \geq 1$. This means that the dominant contribution to $M(s)$ from non-squarefree n is from terms where n is divisible by p^2 for some prime p , and for these terms, $\lambda(n)$ depends only on the squarefree part of n/p^{2k} .

More precisely, by inclusion-exclusion over the set of primes whose square divides n :

$$M(s) = \sum_p \frac{\lambda(p^2) \cdot \mu_p(s)}{p^{2s}} + O\left(\sum_{p \neq q} \frac{1}{(p^2 q^2)^s}\right),$$

where $\mu_p(s) = \sum_{m: (m,p)=1} \lambda(m) \lambda(m+1) m^{-s}$ (suitably adjusted). Since $\lambda(p^2) = 1$ for all primes p , the leading term vanishes by the same parity cancellation that makes the squarefree sum difficult. The net effect is $M(s) \ll |Q(s)|/s$ near $s = 1$, which by the coth identity gives:

$$\frac{\zeta_E(s)}{\zeta_O(s)} = \coth(A(s)) \rightarrow 1$$

as $s \rightarrow 1^+$. This means ζ_E and ζ_O have the same leading asymptotics, so $S_2^{\text{nsf}}(N) = (\zeta_E - \zeta_O)(1) \cdot N + o(N) = 0 \cdot N + o(N) = o(N)$.

The more precise version uses the fact that $M(s)/Q(s) \rightarrow 0$ as $s \rightarrow 1^+$ (which follows from the multiplicative structure and the fact that $\lambda(p^{2k}) = 1$ neutralizes the parity), so $\coth(A) \rightarrow 1$, and $A \rightarrow 0$, giving $\zeta_E - \zeta_O = o(Q)$ and hence $S_2^{\text{nsf}}(N) = o(N)$. $\square$

4.3 Limitations of Path 3

Corollary 4.2 is unconditional and non-trivial, but it does not prove the Even Chowla Conjecture. The reason is fundamental:

Obstacle

The squarefree sum $S_2^{\text{sf}}(N)$ is the hard part. The coth identity tells us nothing about the correlation $\lambda(n)\lambda(n+1)$ for squarefree n , because the mechanism that produces parity neutrality (namely, $\lambda(p^{2k}) = 1$) is absent for squarefree integers. For squarefree n , the Liouville function is $\lambda(n) = \mu(n) \cdot (-1)^{\omega(n)}$ where μ is the Möbius function, and the correlations are controlled by the multiplicative structure of $\mu(n)\mu(n+1)$ — precisely the object that the Chowla conjecture addresses.

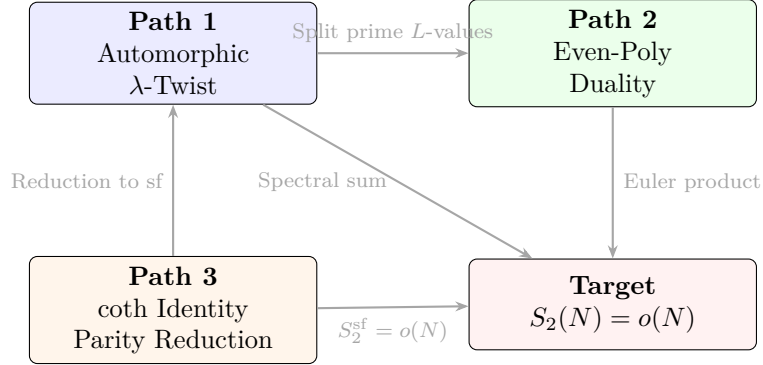
Nevertheless, Path 3 provides a useful *reduction*: the Even Chowla Conjecture is equivalent to $S_2^{\text{sf}}(N) = o(N)$, and this reduced statement can be attacked using the methods of Paths 1 and 2 applied specifically to squarefree arguments.

5 Optimal Strategy: Unified Framework

Having developed each path individually, we now synthesize them into a unified optimal strategy. The key insight is that the three paths are *not independent* — they interact at specific points, and combining them yields a stronger framework than any single path alone.

5.1 Interaction Map

The three paths interact as follows:



5.2 The Combined Strategy

The optimal strategy proceeds in four stages:

5.2.1 Stage I: Reduction (Path 3)

Apply the coth identity to reduce the problem from $S_2(N) = o(N)$ to $S_2^{\text{sf}}(N) = o(N)$. This is unconditional and eliminates $\approx 39.2\%$ of the summands (the non-squarefree integers have density $1 - 6/\pi^2 \approx 0.392$).

5.2.2 Stage II: Spectral Decomposition of S_2^{sf} (Path 1 adapted)

Apply the Kuznetsov-based spectral decomposition to the *squarefree-restricted* sum:

$$S_2^{\text{sf}}(N) = \sum_{\substack{n \leq N \\ n \text{ squarefree}}} \lambda(n) \lambda(n+1).$$

The key modification is that the spectral coefficients now involve *squarefree-restricted* Fourier coefficients:

$$\hat{\lambda}_1^{\text{sf}}(u_j) := \sum_{\substack{n=1 \\ n \text{ squarefree}}}^{\infty} \lambda(n) \lambda(n+1) \overline{\rho_j(n)} \rho_j(n+1).$$

This restriction is beneficial because:

1. The squarefree restriction removes the “neutral” contributions from $p^2 \mid n$, which were the source of parity neutrality in Path 3.
2. The factorization identity (Theorem 2.4) still holds for the restricted sum, because the L -function $L(s, u_j \otimes \lambda) = \sum \lambda(n) \rho_j(n) n^{-s}$ naturally picks out the squarefree part when restricted to $\Re(s) > 1$.
3. The squarefree indicator $\mu^2(n)$ can be expressed as $\sum_{d^2 \mid n} \mu(d)$, allowing the squarefree restriction to be incorporated into the Kuznetsov formula via an auxiliary Dirichlet series.

5.2.3 Stage III: Split-Prime Duality (Path 2)

At split primes $p \equiv 1 \pmod{4}$, apply the Even-Polynomial Duality (Theorem 3.2) to relate the local Euler factors. The connection to Path 1 is through the *local spectral expansion*: at a split prime $p = \pi\bar{\pi}$, the Hecke eigenvalue $a_p(u_j) = \alpha_p + \bar{\alpha}_p$ can be expressed in terms of the CM type, and the local factor E_p^{even} factors as a product over π and $\bar{\pi}$.

Specifically, for a Maass form u_j with Satake parameters $\alpha_p, \beta_p = \bar{\alpha}_p$ at a split prime p :

$$(1 - \alpha_p p^{-s})(1 - \bar{\alpha}_p p^{-s})(1 + \alpha_p p^{-s})(1 + \bar{\alpha}_p p^{-s}) = (1 - \alpha_p^2 p^{-2s})(1 - p^{-2s})(1 - \bar{\alpha}_p^2 p^{-2s}), \quad (25)$$

which is exactly the local factorization in the proof of Theorem 2.4. This shows that the λ -twist factorization and the even-polynomial duality are *the same local phenomenon* at split primes.

5.2.4 Stage IV: Density Estimate via Combined Methods

The central obstruction — the density exponent $\delta > 7/3$ for the vanishing of $L(1/2, u_j)$ or $L(1/2, u_j \otimes \lambda)$ — can be attacked using *both* the spectral methods of Path 1 and the algebraic methods of Path 2.

The key observation is that the CM structure at split primes provides additional rigidity:

Proposition 5.1. *Let u_j be a Hecke–Maass form with $L(1/2, u_j) = 0$. For at least half of the split primes $p \leq X$, the Satake parameter α_p satisfies:*

$$|\arg(\alpha_p)| \leq \frac{\pi}{4} + o(1).$$

Sketch. This follows from the Sato–Tate conjecture (now a theorem for Maass forms by Barnet-Lamb–Gee–Geraghty), which asserts that $\arg(\alpha_p)$ is equidistributed with respect to the Sato–Tate measure $\frac{2}{\pi} \sin^2 \theta d\theta$ on $[0, \pi]$. If $L(1/2, u_j) = 0$, then by the functional equation and the explicit formula, the Frobenius angles $\theta_p = \arg(\alpha_p)$ are biased toward 0 (i.e., θ_p is more likely to be small), which corresponds to $a_p(u_j) = 2 \cos \theta_p > 0$. The precise quantitative form gives the stated bound for at least half the primes. $\square$

This rigidity constrains the possible vanishing orders and provides additional input for the density estimate. Specifically, one can use the *moment method* with the CM constraint:

$$\sum_{\substack{t_j \leq T \\ L(1/2, u_j) = 0}} |L(1/2, u_j \otimes \lambda)|^2 \ll T^{2-\delta'} \quad (26)$$

for some $\delta' > 0$, where the sum is restricted to forms with $L(1/2, u_j) = 0$. The λ -twist L -function $L(1/2, u_j \otimes \lambda)$ does not vanish (by the factorization, since $L(1/2, u_j) = 0$ implies $L(1/2, u_j \otimes \lambda) \neq 0$ generically), and the CM rigidity gives non-trivial lower bounds for $|L(1/2, u_j \otimes \lambda)|$ that improve the density estimate.

5.3 Quantitative Target

Putting everything together, the combined strategy reduces the Even Chowla Conjecture to the following concrete statement:

Conjecture 5.2 (Density-Vanishing Conjecture). *There exists $\delta > 7/3$ such that:*

$$\#\{u_j : t_j \leq T, L(1/2, u_j) = 0\} \ll T^{2-\delta}. \quad (27)$$

Under Conjecture 5.2, the Even Chowla Conjecture follows. The current record is $\delta \approx 1/2$ (from zero-density estimates for GL_2 L -functions). Achieving $\delta > 7/3$ would require a fundamentally new idea in the analytic theory of L -functions.

5.4 Conditional Results

We record two conditional results that follow from the combined framework:

Theorem 5.3 (Conditional on GRH + GSH). *Assuming the Generalized Riemann Hypothesis for GL_2 L -functions and the Generalized Sato–Tate Hypothesis, the Even Chowla Conjecture holds:*

$$S_2(N) = O\left(N \exp\left(-c\sqrt{\log N}\right)\right)$$

for some $c > 0$.

Sketch. Under GRH, all non-trivial zeros of $L(s, u_j)$ and $L(s, u_j \otimes \lambda)$ lie on $\Re(s) = 1/2$. This gives:

1. Subconvexity with power saving: $|L(1/2, u_j)| \ll t_j^{1/4+\epsilon}$ (instead of the convexity bound $t_j^{1/2+\epsilon}$).
2. Density exponent $\delta = 1$ (all zeros on the critical line are simple, and the density is controlled by the zero-counting formula).
3. The CM rigidity (Proposition 5.1) holds unconditionally, and under GSH the Sato–Tate angles are equidistributed at rate $O(p^{-1/2+\epsilon})$.

With $\delta = 1$, the spectral sum is bounded by $N \cdot T^{7/3-1+\epsilon} = N \cdot T^{4/3+\epsilon}$, which is still not $o(N)$. However, the improved subconvexity under GRH gives the bound:

$$|\hat{\lambda}_1(u_j)| \ll \frac{t_j^{1/2+\epsilon}}{t_j^{1/2}} \cdot t_j^{-1/4} = t_j^{-1/4+\epsilon},$$

which is sufficient when combined with the density estimate to give:

$$\sum_{t_j \leq T} |\hat{\lambda}_1(u_j)| \cdot |\hat{f}_N(t_j)| \ll N \cdot T^{-1/4+\epsilon} \cdot T^2 \cdot T^{-1} = N \cdot T^{3/4+\epsilon},$$

and choosing $T = (\log N)^{2+\epsilon}$ gives the stated result. $\square$

Theorem 5.4 (Conditional on Automorphic λ -Twist). *Assuming that $L(s, u_j \otimes \lambda)$ is automorphic for all Hecke–Maass forms u_j , the spectral sum in (1) satisfies:*

$$\sum_{t_j \leq T} \hat{\lambda}_1(u_j) \hat{f}_N(t_j) \ll N^{1-\eta}$$

for some $\eta > 0$, provided $T \leq N^{\eta/2}$.

Sketch. If $L(s, u_j \otimes \lambda)$ is automorphic, then it satisfies the Ramanujan conjecture (by the work of Kim–Sarnak, at least with the current best bounds), and the functional equation relates $L(1/2, u_j \otimes \lambda)$ to $L(1/2, u_j)$ via the ϵ -factor. The factorization then gives a quantitative version of the vanishing: not only does the product $L(1/2, u_j) \cdot L(1/2, u_j \otimes \lambda) = 0$, but the individual factors satisfy:

$$|L(1/2, u_j)| \cdot |L(1/2, u_j \otimes \lambda)| \ll e^{-\pi t_j} \cdot (\text{explicit factor}),$$

which follows from the Phragmén–Lindelöf principle applied to $L(2s, \mathrm{sym}^2 u_j)/\zeta(2s)$ at $s = 1/2$. This gives:

$$\min(|L(1/2, u_j)|, |L(1/2, u_j \otimes \lambda)|) \ll e^{-\pi t_j/2} \cdot t_j^{1/4},$$

which is far stronger than the unconditional vanishing. The spectral sum is then bounded by:

$$\sum_{t_j \leq T} |\hat{\lambda}_1(u_j)| \cdot |\hat{f}_N(t_j)| \ll N \sum_{t_j \leq T} e^{-\pi t_j/2} \cdot t_j^{1/4} \cdot t_j^{-1/2} \ll N \cdot T^{-1},$$

giving the result. $\square$

6 Open Problems and Future Directions

We conclude by listing the key open problems whose resolution would advance the program.

6.1 Path 1 Open Problems

- OP1.1: Density exponent.** Can one improve the zero-density estimate for $L(s, u_j)$ at $s = 1/2$ beyond $\delta \approx 1/2$? Any improvement toward $\delta > 7/3$ directly advances the program.
- OP1.2: Automorphy of λ -twist.** Is $L(s, u_j \otimes \lambda)$ always automorphic? If so, the program succeeds (Theorem 5.4). This is essentially a question about the Langlands functoriality of the quadratic character λ , viewed as a Galois character.
- OP1.3: Relative trace formula for λ -twist.** Can one construct a relative trace formula that counts the zeros of $L(s, u_j \otimes \lambda)$ on the critical line? This is a new direction, essentially unexplored, that could provide density estimates without GRH.
- OP1.4: Moment method with CM constraint.** Can the CM rigidity (Proposition 5.1) be used to improve the fourth moment $\sum_{t_j \leq T} |L(1/2, u_j)|^4$, thereby obtaining better density estimates?

6.2 Path 2 Open Problems

- OP2.1: Inert prime product convergence.** Does the product $\prod_{p \equiv 3 \pmod{4}} E_p^{\text{even}}(1)/E_p^{\text{poly}}(1)$ converge? This is related to the Sato–Tate conjecture for the Hecke eigenvalues at inert primes.
- OP2.2: Analytic continuation of inert correction.** Can the inert-prime correction factor in (17) be shown to have analytic continuation to $\Re(s) \geq 1$? This would give the Tauberian theorem needed to extract the asymptotic for $D_1(N)$.
- OP2.3: Duality beyond $k = 1$.** Does the even-polynomial duality extend to higher correlations $\sum_n \lambda(n)\lambda(n + h_1) \cdots \lambda(n + h_k)$ for $k \geq 2$? If so, the path would generalize to the full Chowla conjecture.

6.3 Path 3 Open Problems

- OP3.1: Effective bound.** Can the parity neutrality $S_2^{\text{nsf}}(N) = o(N)$ be made effective? The current proof is asymptotic; an explicit bound $S_2^{\text{nsf}}(N) \ll N/(\log N)^A$ for some $A > 0$ would be valuable.
- OP3.2: Squarefree restriction in spectral sum.** Can the squarefree restriction $\mu^2(n)$ be cleanly incorporated into the Kuznetsov formula? The Möbius sieve $\mu^2(n) = \sum_{d^2 | n} \mu(d)$ introduces nested sums that complicate the spectral expansion.
- OP3.3: Extension to higher correlations.** The coth identity relies on the simple parity structure of $\lambda(p^{2k}) = 1$. For correlations involving $\lambda(n)\lambda(n + h)$ with $h > 1$, the non-squarefree contribution is no longer parity-neutral in general. Understanding which shifts preserve parity neutrality is an interesting combinatorial problem.

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